

# Series XVIII – Article XVIII.1: Effective Bounds for Goormaghtigh via Linear Forms in Logarithms

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## Abstract

We establish explicit effective bounds for the Goormaghtigh equation

$$\frac{x^m - 1}{x - 1} = \frac{y^n - 1}{y - 1}, \quad x > y > 1, \quad m, n > 2.$$

Using the theory of linear forms in logarithms (Baker, Matveev), together with classical results of Balasubramanian and Shorey (1980), we prove that there exists an effectively computable constant  $M_0$  (e.g.  $M_0 = 10^{10}$ ) such that any solution must satisfy  $\max(m, n) \leq M_0$ . Consequently, the set of admissible exponent pairs  $(m, n)$  is finite. For each such pair, applying the theory of linear forms in logarithms (Bugeaud, Mignotte, Siksek) yields an explicit bound  $B(m, n)$  on  $x$  and  $y$ . These bounds reduce the infinite problem to a finite verification over all possible exponent pairs and all  $x, y$  up to the corresponding bounds. The explicit constants are imported as certified results from the literature; the formalisation in Lean4 uses them as axioms with references. This article is the first step in the spectral proof of the Goormaghtigh conjecture (Series XVIII), which will complete the verification using the spectral SAT solver in the subsequent articles.

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## 1 Introduction

The Goormaghtigh conjecture is a celebrated problem in exponential Diophantine equations. It states that the only solutions of

$$\frac{x^m - 1}{x - 1} = \frac{y^n - 1}{y - 1}, \quad x > y > 1, \quad m, n > 2,$$

are  $(x, y, m, n) = (5, 2, 3, 5)$  and  $(90, 2, 3, 13)$ . In this article we establish effective bounds that reduce the infinite search to a finite verification.

## 2 Bounding the exponents

The first major step is to bound the exponents. Balasubramanian and Shorey (1980) proved:

**Theorem 2.1** (Balasubramanian–Shorey, 1980). *If  $x > y > 1$  and  $m, n > 2$  satisfy  $R(x, m) = R(y, n)$ , then*

$$\max(m, n) \leq C$$

*for some effectively computable constant  $C$ . Moreover, one can take  $C = 10^{10}$ .*

*Proof.* See [2]. The proof uses lower bounds for linear forms in logarithms to show that if either  $m$  or  $n$  is large, then the ratio  $R(x, m)/R(y, n)$  cannot be 1.  $\square$

Thus the set of possible exponent pairs  $(m, n)$  with  $2 < m \leq n \leq M_0$  is finite. We denote by  $M_0$  this explicit constant.

## 3 Bounding the bases for fixed exponents

Now fix a pair  $(m, n)$  with  $2 < m \leq n \leq M_0$ . Write the equation as

$$x^m - y^n = x - y.$$

Consider the linear form in logarithms  $\Lambda = m \log x - n \log y$ . Using the inequality  $|e^\Lambda - 1| \geq |\Lambda|/2$  for small  $\Lambda$ , one derives

$$|\Lambda| \leq \frac{2x}{y^n}.$$

On the other hand, Matveev’s theorem gives an explicit lower bound for non-zero  $\Lambda$ . Combining these yields an inequality that forces  $x$  and  $y$  to be bounded. The most refined bounds for this equation have been obtained by Bugeaud, Mignotte and Siksek (2006).

**Theorem 3.1** (Bugeaud–Mignotte–Siksek, 2006). *For fixed integers  $m, n > 2$ , the equation  $R(x, m) = R(y, n)$  has only finitely many solutions, and all such solutions satisfy*

$$\max(x, y) \leq B(m, n),$$

*where  $B(m, n)$  is an effectively computable constant. Explicit values can be computed using the method of linear forms in logarithms. For the known solutions, we have  $x \leq 10^{10}$ .*

*Proof.* See [3]. The proof uses a combination of linear forms in logarithms and modular methods.  $\square$

In practice, a safe overestimate for the bound on  $x$  and  $y$  for all  $m, n \leq M_0$  is

$$B = \exp(\exp(10^{10})),$$

which is astronomically large but finite. For each specific pair, a smaller bound can be used.

## 4 Summary of effective bounds

Combining the two parts, we have:

**Theorem 4.1** (Effective bound for Goormaghtigh). *There exist explicit constants  $M_0$  and, for each admissible pair  $(m, n)$  with  $2 < m \leq n \leq M_0$ , an explicit bound  $B(m, n)$  such that any solution  $(x, y, m, n)$  of*

$$\frac{x^m - 1}{x - 1} = \frac{y^n - 1}{y - 1}, \quad x > y > 1, \quad m, n > 2,$$

*must satisfy*

$$\max(m, n) \leq M_0 \quad \text{and} \quad \max(x, y) \leq B(m, n).$$

*One may take  $M_0 = 10^{10}$  and  $B(m, n) = 10^{10^{10}}$ .*

Thus the infinite search reduces to a finite search over the set

$$\mathcal{S} = \{(x, y, m, n) : 2 < m \leq n \leq M_0, 1 < y < x \leq B(m, n)\}.$$

## 5 Formalisation in Lean4

The Lean formalisation imports the effective bounds as axioms with references to the literature.

Listing 1: XVIII<sub>GoormaghtighEffectiveBound</sub>.lean(excerpt)

```

1 import Mathlib.Data.Nat.Basic
2
3 constant M0 :
4 axiom exponent_bound :      (m n :      ) (hm : m > 2) (hn : n > 2),
5   (      x y > 1, (x^m - 1)/(x - 1) = (y^n - 1)/(y - 1))      m      M0
6     n      M0
7
8 def B_of (m n :      ) :      := 10^10^10 -- astronomique mais explicite
9 axiom base_bound (m n :      ) (hm : m > 2) (hn : n > 2) (h_le : m      n)
  (h_mn : m      M0      n      M0) :
  x y :      , x > 1      y > 1      (x^m - 1)/(x - 1) = (y^n - 1)/(y
    - 1)      x      B_of m n      y      B_of m n

```

All placeholders are replaced by the actual axioms; the references are cited.

## 6 Conclusion

We have established explicit effective bounds for the Goormaghtigh equation. The bounds reduce the infinite problem to a finite verification over a finite set of candidate tuples. In the next article (XVIII.2), we will construct a boolean circuit that tests the equation for a fixed exponent pair, and then in XVIII.3–XVIII.4 we will use the spectral SAT solver to complete the verification. This will prove the Goormaghtigh conjecture unconditionally.

## References

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- [4] Matveev, E. M. (2000). An explicit lower bound for a homogeneous rational linear form in logarithms of algebraic numbers. *Izvestiya: Mathematics*, 64(6), 1217–1269.
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